A SINGULAR GENUS 0 PROPER CURVE WITH H^0(X, O_X) = K IS P^1 OVER A NONTRIVIAL FINITE EXTENSION GLUED TO SPEC K AT ONE RATIONAL POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the normalization sequence, the conductor ideal with its defining exact sequence, the commutative-subalgebra dimension bound, the descent I = nu_*I', the estimate dim H^0(X', A') <= 2n-2, the divisibility argument, and the final identification of both rational points and the structure sequence are all retained verbatim.

Editorial note. The label lemma-genus-zero-singular names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Normalization of a singular curve. Varieties, Sections varieties-section-normalization and varieties-section-normalization-one-dimensional, with Lemma varieties-lemma-regular-functions-proper-variety. Supplies the finite normalization nu : X' -> X, the finite extension k' = H^0(X', O_{X'}) and the short exact sequence 0 -> O_X -> nu_*O_{X'} -> Q -> 0 supported in finitely many points.
- Classification of genus 0 curves without H^0 correction. Same chapter, Proposition proposition-projective-line. Identifies X' as P^1_{k'} once H^1(X', O_{X'}) = 0 and the conductor quotient is trivial on X'.

Source reference. Original lemma 01SB: sources/stacks-chow/tags.txt, source lines 2332–2332. Prerequisite pointer invoked at proof line(s) 2301; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C6U: sources/stacks-curves/pinned-original-source.tex, source lines 2138–2138. Prerequisite pointer invoked at proof line(s) 2320; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C6P: sources/stacks-chow/tags.txt, source lines 15304–15304. Prerequisite pointer invoked at proof line(s) 2316; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04L2: sources/stacks-chow/tags.txt, source lines 5993–5993. Prerequisite pointer invoked at proof line(s) 2251; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0BXQ: sources/stacks-chow/tags.txt, source lines 14988–14988. Prerequisite pointer invoked at proof line(s) 2247; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0C44: sources/stacks-chow/tags.txt, source lines 15214–15214. Prerequisite pointer invoked at proof line(s) 2248; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a proper curve over a field k with $H^0(X, \mathcal{O}_X) = k$. Assume X is singular and has genus 0. Then there exists a diagram*

$$\begin{array}{ccccc} x' & \longrightarrow & X' & \longrightarrow & \mathrm{Spec}(k') \\ \downarrow & & \downarrow{\scriptstyle \nu} & & \downarrow \\ x & \longrightarrow & X & \longrightarrow & \mathrm{Spec}(k) \end{array}$$

where

(1) *k'/k is a nontrivial finite extension,*
(2) *$X' \cong \mathbf{P}^1_{k'}$,*
(3) *x' is a k'-rational point of X',*
(4) *x is a k-rational point of X,*
(5) *$X' \setminus \{x'\} \to X \setminus \{x\}$ is an isomorphism,*
(6) *$0 \to \mathcal{O}_X \to \nu_*\mathcal{O}_{X'} \to k'/k \to 0$ is a short exact sequence where $k'/k = \kappa(x')/\kappa(x)$ indicates the skyscraper sheaf on the point x.*

Proof. Let $\nu : X' \to X$ be the normalization of X, see Varieties, Sections 0BXQ and 0C44. Since X is singular ν is not an isomorphism. Then $k' = H^0(X', \mathcal{O}_{X'})$ is a finite extension of k (Varieties, Lemma 04L2). The short exact sequence

$$0 \to \mathcal{O}_X \to \nu_*\mathcal{O}_{X'} \to \mathcal{Q} \to 0$$

and the fact that $\mathcal{Q}$ is supported in finitely many closed points give us that

(1) $H^1(X', \mathcal{O}_{X'}) = 0$, i.e., X' has genus 0 as a curve over k',
(2) there is a short exact sequence $0 \to k \to k' \to H^0(X, \mathcal{Q}) \to 0$.

In particular k'/k is a nontrivial extension.

Next, we consider what is often called the *conductor ideal*

$$\mathcal{I} = \mathcal{H}om_{\mathcal{O}_X}(\nu_*\mathcal{O}_{X'}, \mathcal{O}_X)$$

This is a quasi-coherent $\mathcal{O}_X$-module. We view $\mathcal{I}$ as an ideal in $\mathcal{O}_X$ via the map $\varphi \mapsto \varphi(1)$. Thus $\mathcal{I}(U)$ is the set of $f \in \mathcal{O}_X(U)$ such that $f\,(\nu_*\mathcal{O}_{X'}(U)) \subset \mathcal{O}_X(U)$. In other words, the condition is that f annihilates $\mathcal{Q}$. In other words, there is a defining exact sequence

$$0 \to \mathcal{I} \to \mathcal{O}_X \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{Q}, \mathcal{Q})$$

Let $U \subset X$ be an affine open containing the support of $\mathcal{Q}$. Then $V = \mathcal{Q}(U) = H^0(X, \mathcal{Q})$ is a k-vector space of dimension $n-1$. The image of $\mathcal{O}_X(U) \to \mathrm{Hom}_k(V, V)$ is a commutative subalgebra, hence has dimension $\leq n-1$ over k (this is a property

of commutative subalgebras of matrix algebras; details omitted). We conclude that we have a short exact sequence

$$0 \to \mathcal{I} \to \mathcal{O}_X \to \mathcal{A} \to 0$$

where $\text{Supp}(\mathcal{A}) = \text{Supp}(\mathcal{Q})$ and $\dim_k H^0(X, \mathcal{A}) \leq n - 1$. On the other hand, the description $\mathcal{I} = \mathcal{H}om_{\mathcal{O}_X}(\nu_*\mathcal{O}_{X'}, \mathcal{O}_X)$ provides $\mathcal{I}$ with a $\nu_*\mathcal{O}_{X'}$-module structure such that the inclusion map $\mathcal{I} \to \nu_*\mathcal{O}_{X'}$ is a $\nu_*\mathcal{O}_{X'}$-module map. We conclude that $\mathcal{I} = \nu_*\mathcal{I}'$ for some quasi-coherent sheaf of ideals $\mathcal{I}' \subset \mathcal{O}_{X'}$, see Morphisms, Lemma 01SB. Define $\mathcal{A}'$ as the cokernel:

$$0 \to \mathcal{I}' \to \mathcal{O}_{X'} \to \mathcal{A}' \to 0$$

Combining the exact sequences so far we obtain a short exact sequence $0 \to \mathcal{A} \to \nu_*\mathcal{A}' \to \mathcal{Q} \to 0$. Using the estimate above, combined with $\dim_k H^0(X, \mathcal{Q}) = n - 1$, gives

$$\dim_k H^0(X', \mathcal{A}') = \dim_k H^0(X, \mathcal{A}) + \dim_k H^0(X, \mathcal{Q}) \leq 2n - 2$$

However, since X' is a curve over k' we see that the left hand side is divisible by n (Varieties, Lemma 0C6P). As $\mathcal{A}$ and $\mathcal{A}'$ cannot be zero, we conclude that $\dim_k H^0(X', \mathcal{A}') = n$ which means that $\mathcal{I}'$ is the ideal sheaf of a k'-rational point x'. By Proposition 0C6U we find $X' \cong \mathbf{P}^1_{k'}$. Going back to the equalities above, we conclude that $\dim_k H^0(X, \mathcal{A}) = 1$. This means that $\mathcal{I}$ is the ideal sheaf of a k-rational point x. Then $\mathcal{A} = \kappa(x) = k$ and $\mathcal{A}' = \kappa(x') = k'$ as skyscraper sheaves. Comparing the exact sequences given above, this immediately implies the result on structure sheaves as stated in the lemma. □